

Your first analysis

NSDF-Data student guide 2 — Linux edition

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Tip

What you will do. Download one set of real detector data, open it in a notebook, look at it, and reproduce one number from an existing note. You will need the setup guide for Linux finished first.

Before you start

- **Time:** about two hours.
- **You need:** the setup guide finished (your environment `darkmatter_cli_env` exists and works).
- **Check:** open a terminal (Step 1 of the setup guide), and run `conda activate darkmatter_cli_env`. The start of the line should show (`darkmatter_cli_env`).

Words you will meet

Word	What it means here
Series	One data-taking session, named like 07221203_2025.
Dump	One file set taken from a series, numbered F0001, F0002, ...
Event	One trigger of the detector. Events are numbered in time order.
Detector / channel	One physical detector, and one of its four sensor channels. We use detector 0, channel 0.
Trace	The recorded signal for one channel in one event: 4096 numbers, one per sample.
Sample	One reading. Each sample is 1.6 microseconds long.
ADC counts	The unit of the numbers in a trace.
Baseline	The flat level a trace sits at when nothing happens.

Step 1. Read three notes first (about 30 minutes)

Open the notes site at villano-lab.github.io/NSDF-Data and read these, in order:

1. **Note 1**, *Run76 series catalog*: what a series and a dump are.
2. **Note 1a**, *python library structure*: what each part of the code does.
3. **Note 2**, *Noise-pulse selection*: how we pick quiet traces.

You do **not** need to follow every number. Just learn the words in the table above.

Step 2. Download the data

First go to your home folder, so the data lands in the right place. Type this, then press Enter:

```
cd ~
```

Now download the data:

```
nsdf-cli download 07221203_2025_F0001
```

The download puts the data in an `idx` folder **in the folder you are in**. Because you went to your home folder first, it lands in `~/idx`, inside a folder named `07221203_2025_F0001`. The data is about 34 MB. Wait until the prompt comes back.

Careful

Never download from inside the NSDF-Data folder. Its `idx` folder would sit inside the project, where Git can pick it up. If you have already downloaded there, move the `idx` folder to your home folder, and tell your lead.

Careful

Do not move or edit the downloaded files, and do not add them to Git. They are not part of the project.

Tip

If the message says the folder already exists, the data may already be there. Ask the project lead before you delete anything.

Step 3. Open Jupyter in the right folder

Notebooks must be opened from the folder `R76/analysis_notes`, inside the project. In the terminal, type these lines:

```
conda activate darkmatter_cli_env
cd ~/Research/NSDF-Data/R76/analysis_notes
jupyter lab
```

Your web browser opens a Jupyter page. Click **File**, then **New**, then **Notebook**. If it asks which kernel to use, choose **darkmatter_cli_env**.

Tip

A notebook is a list of boxes called **cells**. Type code in a cell and press **Shift + Enter** to run it. The result appears under the cell. Press **Enter** (without Shift) to keep typing in the same cell.

Give your notebook a name: click the name at the top, type first-analysis-yourname, and press Enter.

Step 4. Load the data and count the traces

Copy this into the first cell and press **Shift + Enter**:

```
import sys
from pathlib import Path

# Tell Python where the pulse library (the python folder) is.
# This notebook is in R76/analysis_notes, so it is two folders up.
sys.path.insert(0, str(Path.cwd().parents[1] / "python"))

from nsdf_dark_matter.idx import load_all_data
import pulse_io as pio

# Open the dump you downloaded.
cdms = load_all_data(str(Path.home() / "idx" / "07221203_2025_F0001"))

# Take every trace from detector 0, channel 0, into one table.
ids, pulses = pio.load_channel_batch(
    cdms, pio.filter_by_detector(cdms.get_detector_ids(), 0), 0)

print(pulses.shape)
```

Checkpoint

The output is (1517, 4096): 1517 traces, each with 4096 samples. If you get a different number, stop and check the dump name and the folder in Step 2.

Step 5. Look at a trace

Put this in a new cell (click + in the toolbar, or use the menu **Insert**, then **Insert Cell Below**), and run it:

```
import matplotlib.pyplot as plt

plt.plot(pulses[0])
plt.xlabel("sample (1.6 microseconds each)")
plt.ylabel("ADC counts")
plt.show()
```

Now change the 0 to 1, 2, 3 and run again. Most traces look like flat noise, with a little jitter.

Step 6. Reproduce a number from Note 2

Note 2 says that for this dump, a strict selection leaves **179 quiet traces**. Run this in a new cell:

```
from dataclasses import replace
from pulse_config import DEFAULT_CONFIG
import pulse_quantities as pq
import pulse_cuts as pc
import numpy as np

c500 = replace(DEFAULT_CONFIG, pretrigger_samples=500)
quiet = pc.excursion_band(pulses, c500) & (np.log10(pq.bstd(pulses)) < 0.5)
print(quiet.sum())
```

Checkpoint

The output is **179**. If you see a different number, that is worth knowing. Write down the number you got and the steps you took, and tell the project lead. Do not change the project's code to make the number match.

Step 7. Write down what you did

In Jupyter, click **+** to add a cell and change its type from *Code* to **Markdown** in the toolbar. In that cell, write a few sentences in plain English: what you loaded, what you plotted, and what number you got. Then press **Shift + Enter**.

Save with **Command + S** (macOS) or **Ctrl + S** (Windows and Linux).

Once your branch exists (setup guide, Step 8), save your notebook to Git:

```
cd ~/Research/NSDF-Data
git add R76/analysis_notes/first-analysis-yourname.ipynb
git commit -m "First analysis: load dump 1 and count the quiet traces"
git push -u origin student-yourname
```

Careful

Commit only your notebook. Check the list before committing with `git status`. If you see `.bin` files or anything from the `idx` folder, do not add them.

You are done when

- ☐ (1517, 4096) printed after loading the data.
- ☐ You plotted at least three traces and can say what the flat ones look like.
- ☐ You got 179 from the selection, or wrote down what you got and why you think it differs.
- ☐ Your notebook is saved, and committed to your branch if you have access.

Common problems

What you see	What to do
ModuleNotFoundError: No module named 'nsdf_dark_matter'	You are not in the environment. Run <code>conda activate darkmatter_cli_env</code> , then restart the notebook kernel (Kernel , then Restart Kernel).
ModuleNotFoundError: No module named 'pulse_io'	Jupyter was started from the wrong folder. Close it and start it again from R76/analysis_notes.
FileNotFoundError on the dump folder	The download did not finish, or it went somewhere else. Check the folder in Step 2.
A huge spike at the very start of traces	This is a real electronics glitch in the first few samples, not a bug. It is explained in Note 2a.
The kernel is not darkmatter_cli_env	Click the kernel name at the top right and choose darkmatter_cli_env .

Careful

Do not run `07221203_2025_dump1_noise.ipynb` unless the project lead says it is fine. It overwrites a shared archive file.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

When you write, include the step number, the command you typed, and the last few lines the computer printed. A screenshot helps. Nobody will be annoyed by a question about a step that did not work.

Session Info

Guide version 4, 6 October 2026, Linux edition. Written for beginners; please tell the project lead where a step was unclear.

Key links

- Note 1: <https://villano-lab.github.io/NSDF-Data/notes/note-01-run76-series-catalog.html>
- Note 1a: <https://villano-lab.github.io/NSDF-Data/notes/note-01a-pulse-library-structure.html>
- Note 2: <https://villano-lab.github.io/NSDF-Data/notes/note-02-noise-selection.html>
- Repository: <https://github.com/villano-lab/NSDF-Data>